

FinBERT Under Geopolitical Domain Shift: Evaluating LoRA and Fine-Tuning for Cross-Domain Sentiment

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Abstract

This project examines how adaptation capacity affects FinBERT's ability to generalize to geopolitical sentiment classification under domain shift. I compare three configurations: Low Rank Adaptations LoRA ($r = 4$), LoRA ($r = 16$), and full fine-tuning, trained on Financial PhraseBank (~2,260 sentences) and evaluated on a curated dataset of 178 geopolitical news headlines (2018–2026). A zero-shot FinBERT baseline achieves macro-F1 score 0.49 on geopolitical data. In-domain performance improves with adaptation capacity, with LoRA $r = 4$ achieving F1 score 0.66, LoRA to full fine-tuning achieving 0.955. These gains do not transfer out-of-domain. The full fine-tuned model drops to macro-F1 = 0.37 on geopolitical headlines, below the zero-shot baseline, experiencing negative transfer. All models exhibit a strong bias toward neutral predictions under domain shift, suggesting that geopolitical sentiment is expressed differently from financial sentiment. These results show that increasing adaptation capacity improves in-domain performance but does not improve and may degrade cross-domain generalization.

GitHub Repository: <https://github.com/lambchops27-coder/stat359>

1. Introduction

1.1 Motivation

Large language models fine-tuned on domain-specific corpora have demonstrated strong in-distribution performance across financial NLP tasks. FinBERT, a financial-domain adaptation of BERT, represents the current standard for financial sentiment analysis. Nonetheless, recent geopolitical developments, such as trade wars, sanctions, technological export controls and strategic decoupling introduces a category of market-moving language that differs structurally and lexically from traditional financial reporting. This increasing influence on firm operations and plans, investor confidence, and thereby financial market behaviour. This phenomenon raises the importance of accurately interpreting geopolitical news.

Furthermore, geopolitical language often differs in tone and structure from conventional financial reporting that FinBERT is trained on. Differences in linguistic structure may limit effectiveness of models trained solely on financial data. Understanding how effective domain-specific LLMs generalize under this domain shift can bring fruitful improvements for risk analysis in finance, given the increasing influence of geopolitical events.

1.2 Problem Statement

This project investigates the extent to which a financial-domain language model, FinBERT, can generalize to sentiment classification on geopolitical news headlines. Specifically, focus is placed on examining whether improving in-domain performance through fine-tuning or parameter-efficient adaptation would result in an improved out-of-domain performance under geopolitical domain shift.

1.3 Research Questions

1. How does increasing adaptation capacity (LoRA rank and full fine-tuning) affect in-domain sentiment classification performance on Financial PhraseBank?
2. Does improved in-domain performance translate to better sentiment classification on geopolitical news headlines?
3. How do different adaptation methods affect model behavior under domain shift, particularly with respect to class prediction bias?

1.4 Summary of Contributions

- Implemented and compared three adaptation strategies for FinBERT, namely LoRA ($r=4$), LoRA ($r=16$) and full fine-tuning
- Constructed a curated dataset of 178 geopolitical news headlines with sentiment labels

- Evaluated model performance both in-domain via Financial PhraseBank texts and out-of-domain via geopolitical news headline text
- Conducted an analysis on training dynamics using loss curves and macro-F1 metrics
- Identified systematic neutral prediction bias under geopolitical domain shift

2. Background and Related Work

Hu et al. (2021) introduces LoRA, a method constraining transformer weight updates to low-rank matrix products inserted in attention layers, which reduce trainable parameters by orders of magnitude whilst also keeping competitive task performance. This project implements this framework by targeting Query and Value attention matrices over all 12 FinBERT layers. The framework is extended by answering the unaddressed question of examining whether rank selection influences out-of-distribution generalization rather than in-domain performance.

Malo et al. (2014) introduce Financial PhraseBank, a financial sentiment corpus at the sentence-level annotated by domain experts, with the sentences_allagree split providing 2,264 high-confidence examples across three sentiment classes. This project uses Financial PhraseBank as the training corpus and adopts their annotation methodology, where sentiment is anchored to implied market impact, instead of subjective tone, as the basis for the geopolitical test set labeling rubric.

Lastly, Guo et al. (2025) fine-tune and explain FinBERT on standard financial text, achieving zero-shot macro-F1 score 0.555 and fine-tuned macro-F1 score 0.707. These findings establish the performance reference point against this project's zero-shot macro-F1 = 0.49 on geopolitical headlines, thereby quantifying the domain-shift penalty. This project deviates from their setup by evaluating out-of-distribution geopolitical text rather than in-distribution financial benchmarks.

3. Methodology and Experimental Setup

3.1 Data

3.1.1 Source and Characteristics of Training Data

The Financial Phrasebank (FP) dataset by takala served as training data for both the baseline and LoRA-engineered models. The FP dataset contained 2,260 sentences, each labeled by multiple human annotators that come to unanimous agreement on the sentence's sentiment. Sentences in the FP dataset were classed as either Positive, Negative or Neutral.

The FP dataset was split 80/20 into training (1,811 sentences) and validation (453 sentences) sets via a fixed random seed of 42 to ensure reproducibility.

3.1.2 Source and Characteristics of Testing Data

A data-scraping approach was implemented to gather geopolitical news headlines to analyze FinBERT's sentiment classification performance under a domain shift towards geopolitical text. Again, news headlines are classified under Positive, Negative or Neutral sentiment, following the standard benchmark for financial sentiment classification.

A dedicated data-scraping program, [scraping.py](#) was written as a filter for temporal and thematic diversity amongst scraped headlines. Filters included specification of three time periods: 2018-2021, 2022-2023 and 2024-2026. A list of key words associated with US-China technological competition, trade wars, supply chain decoupling or sanctions were incorporated to ensure scraped headlines aligned with the rising geopolitical atmospheres concerning heightened geopolitical securitization, supply chain decoupling.

Following scraping, headlines were manually labeled adhering to a four-rule rubric, requiring: (1) involvement of a major economic or political power, (2) an identifiable economic transmission mechanism or impact to US-linked markets, (3) a reported event, as opposed to an opinion or speculation, (4) macro-level rather than firm-specific.

Headlines failing any one of the rules were excluded as irrelevant. Remaining headlines were labeled based on implied directional market impact. The final label distribution was 57% Negative, 25% Neutral, 17% Positive.

3.2 Model Architecture

The baseline model used was FinBERT. Trained using a diverse collection of financial documents between 2009 to 2016, FinBERT specializes in sentiment classification when given finance-related information such as firm earning reports, company fundamentals or macroeconomic data summaries. The model was imported from HuggingFace.

3.3 Techniques

I fine-tuned the FinBERT model using two parameter-efficient configurations (LoRA), and one full fine-tuning baseline. Details of changes implemented are enlisted in Figure 1 below.

Configuration	Changes involved
LoRA (r = 4)	Low-rank adaptation applied to attention matrices across all transformer layers. Approximately 300K trainable parameters out of 110M total.
LoRA (r = 16)	Higher-rank adaptation, same target modules. Approximately 1.2M trainable parameters.
Full fine-tuning	All 110M parameters updated. Standard supervised fine-tuning baseline.

Figure 1:

Hyperparameter Choices

FinBERT was fine-tuned across 3 configurations with consistent hyperparameters: learning rate $2e-5$, training batch size = 16, evaluation batch size 32, maximum sequence length 128 tokens, and 5 epochs with seed 42. For LoRA configurations, adapters were applied to Query and Value attention matrices across all 12 layers.

LoRA $r = 4$ used $\alpha=8$ with $\sim 300K$ trainable parameters, LoRA $r = 16$ used $\alpha = 32$, $\sim 1.2M$ trainable parameters. α/r was held constant across both LoRA configs to isolate rank as the sole variable. The best checkpoint was selected using `load_best_model_at_end = True` with macro-F1 as selection criteria.

3.4 Evaluation Metrics

Three core metrics listed in Figure 2 are used to evaluate sentiment classification performance of each model under a geopolitical domain shift.

Metric	Purpose
Macro-F1 Scores	Equal weights all sentiment classes and accounts for class imbalance.
Confusion Matrix	Examination of class-level prediction behavior, especially under domain shift
Training Dynamics	Evaluated via loss curves (training and validation) and Macro-F1 scores across each epoch to assess convergence and overfitting.

Figure 2

Each model's domain shift performance is compared to their in-domain performance. This approach is adopted because it standardizes comparison, where performance in a geopolitical domain is relative to the model's in-domain performance of the domain it specializes in.

4. Results

4.1 In-Domain Performance (Financial Phrasebank)

All fine tuned models in Figure 3 show strong performance in sentiment classification on the FP dataset. When increasing adaptations via raising LoRA ranks, macro F1 scores see clear improvements subsequently.

The LoRA $r=4$ model achieved a validation macro-F1 score of approximately 0.66. LoRA $r = 16$ saw improvements in performance with a macro F1-score of 0.89. Full fine-tuning yields the highest performance, reaching macro-F1 scores above 0.94.

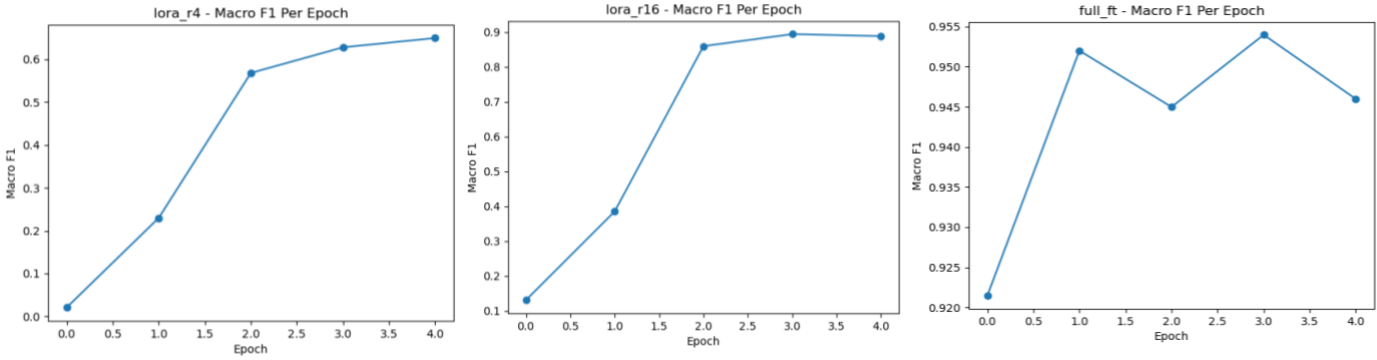


Figure 3: Macro F1 score comparison across epochs per model

These results suggest that increasing model capacity, through higher LoRA rank or full parameter updates enables the model to better capture sentiment patterns in financial text. The improvement from LoRA $r=4$ to $r=16$ suggests that low-rank adapters with limited capacity may underfit the task, while higher-rank adapters can represent richer task-specific features.

4.2 Training Dynamics

Training and validation loss curves demonstrate a consistent convergence across all configurations. Seen in Figure 4, in all cases, training loss decreases steadily over epochs with no significant oscillations or divergences observed. Additionally, validation loss decreases sharply within the first few epochs, then stabilizes, indicating the models learn the FP classification task fast. This is consistent with the relatively small dataset size and usage of pre-trained language model.

Full fine-tuning exhibited the fastest convergence and low final loss values, which reflects higher flexibility when updating model parameters. LoRA models converge more gradually, but still achieve stable training behavior.

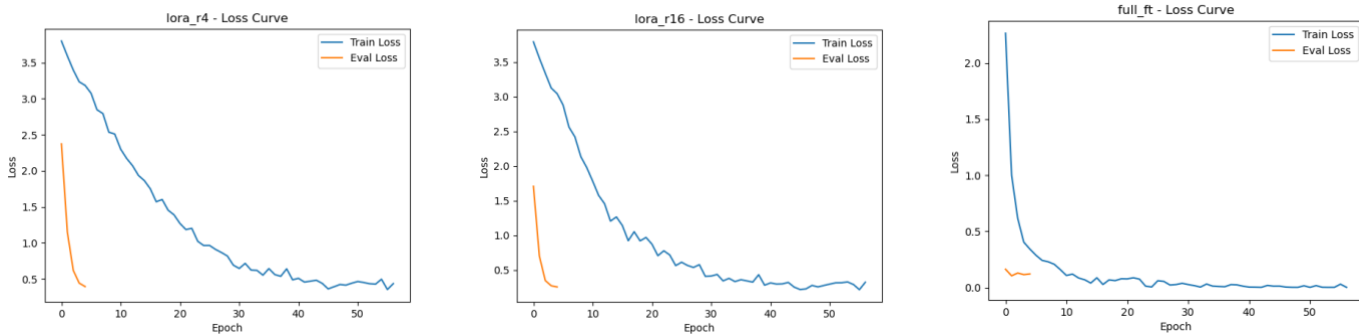


Figure 4: Loss curve comparison across epochs per model

4.3 Zero-Shot Performance on Geopolitical Headlines

The zero-shot FinBERT baseline achieves a macro-F1 score of approximately 0.49 on the geopolitical headline dataset. Performance varies across classes, with relatively good performance on negative sentiment ($F1 = 0.62$), but poorly on positive (0.36) and neutral (0.45) classes.

4.4 Out of Domain Performance After Fine-Tuning

Despite strong in-domain improvements, fine-tuned models do not echo corresponding gains on geopolitical headline classification.

Confusion matrices seen in Figure 4 for LoRA models $r=4$ and $r=16$ show strong bias towards predicting Neutral class. One example is the majority of negative headlines misclassified as Neutral, with LoRA models predicting neutral for over 90% of certain classes.

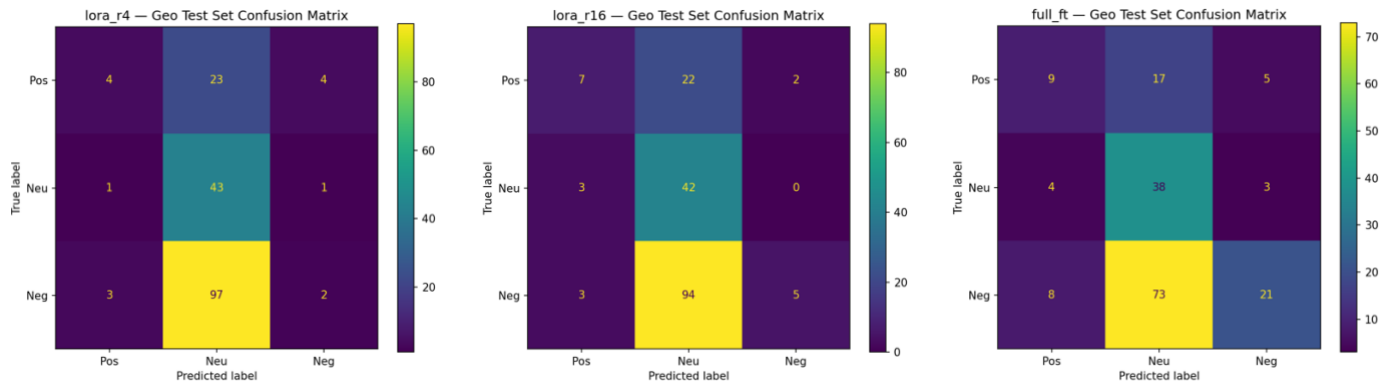


Figure 6: Confusion matrix per model

The issue of Neutral-leaning misclassification does not change when increasing LoRA rank to 16. Full fine-tuning shows some improvement relative to LoRA models. Specifically there is slightly better class separation and reduced neutral bias. However, the full fine-tuned model still fails to reliably classify geopolitical sentiment, especially for positive and negative headlines expressed in indirect or diplomatic language.

4.5 Qualitative Analysis

Qualitative examples further illustrate the limitations of all models under domain shift. For instance, clearly negative headlines involving tariffs or sanctions are frequently classified as neutral, and positive events such as diplomatic agreements are also misclassified.

This consistent pattern suggests that geopolitical sentiment is often expressed in less explicit terms than financial sentiment, making it more difficult for models trained on financial corpora to interpret correctly.

5 Discussion & Error Analysis

5.1 Interpretation

5.1.1 In-Domain Performance (Financial Phrasebank)

The in-domain performance suggests that increasing model capacity, through higher LoRA rank or full parameter updates enables the model to better capture sentiment patterns in financial text. The improvement from LoRA $r=4$ to $r=16$ suggests that low-rank adapters with limited capacity may underfit the task, while higher-rank adapters can represent richer task-specific features.

5.1.2 Zero-Shot Performance on Geopolitical Headlines

Macro-F1 score from zero-shot baseline indicates that FinBERT can detect overtly negative events, likely due to overlap with financial risk language, it also reveals struggles to interpret more nuanced or indirect sentiment, particularly in cases of geopolitical de-escalation or neutral reporting.

5.1.3 Out of Domain Performance

The limited improvement in performance seen in a strong neutral-leaning classification model confusion matrices for each indicates that additional adaptation capacity improves in-domain fit but does not translate into better generalization under domain shift.

6. Limitations

6.1 Small and imbalance distribution of test dataset

The geopolitical headline dataset contains 178 headlines with a skewed distribution towards negative sentiment (57% negative headlines). The positive class only contains 31 examples, which makes per-class F1 estimates statistically noisy. A misclassification of 3-4 correct predictions can meaningfully change Positive F1 score. Hence, results must be interpreted as indicative instead of definitive, especially for per-class comparisons.

6.2 Domain and structural mismatch

Sentences in FP on average substantially more tokens than news headlines and uses clear, evaluative language. Contrarily, geopolitical headlines are much shorter, structurally compressed and express sentiment indirectly. This structural mismatch compounds the lexical domain shift.

6.3 LoRA evaluation caveat

LoRA results for geopolitical headline classification must be interpreted with caution. Adapter weights did not load correctly during evaluation, as confirmed by the load report showing zero adapter parameters applied. This is a known PEFT initialisation issue. This affects validity of out-of-domain LoRA evaluation since weights failed to apply during inference. Hence, LoRA geopolitical results must be interpreted with caution and cannot be directly comparable to full fine-tuning.

6.4 Temporal, source and labeling bias

Test set skews toward 2024–2026 due to API weighting towards recent publications. Guardian (center left) and Reuters (center) are the only sources, where there were no right-leaning financial outlets included. Additionally, tariffs and trade dominate themes (62%), underrepresenting chip war and critical minerals topics that most directly probe securitisation vocabulary. Headline labelling also misses inter-annotator validation, where I serve as the only annotator for headline sentiment. This approach may introduce subjectivity bias which is suboptimal in geopolitical sentiment classification.

7. Failure Modes

All models demonstrate a neutral-class collapse on geopolitical text. Specifically the fine tuned models learn that sentences without clear, explicit financial sentiment markers (e.g. gains, losses) are neutral. While this is a valid heuristic on the FP, it is catastrophic on geopolitical text given that sentiment is expressed via diplomatic framing, conditional threats, and geopolitical escalation vocabulary absent from the training distribution. The zero-shot baseline model, not trained to associate neutral with the absence of financial markers, actually performs better as it held broader contextual understanding.

8. Conclusion

Increasing FinBERT's adaptation capacity can improve in-domain financial sentiment performance, but cannot be generalized to the domain of geopolitics. Full fine-tuning achieves near-ceiling results on the FP dataset with macro-F1 score 0.955, yet performs worse than zero-shot baseline on geopolitical headlines (0.37 vs 0.49). These numbers are consistent with negative transfer, and may indicate catastrophic forgetting. All four models also show neutral-class collapse under domain shift, indicating that geopolitical sentiment is structurally different from financial language. Robust performance hence likely requires domain-relevant training data rather than stronger fine-tuning.

9. Contributions

This project was completed individually. All components from dataset curation, labeling, scraping pipeline, model training, evaluation pipeline, and analysis were completed by Fung Lok Lam.

10. References

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